

Efficient Structure-Aware Discrete Clustering via Multi-Order Anchor Graphs

Ben Yang^{ID}, *Member, IEEE*, Xuetao Zhang, *Member, IEEE*, Yu Zhou^{ID}, Haoxin Wu^{ID},
Feiping Nie^{ID}, *Senior Member, IEEE*, and Badong Chen, *Senior Member, IEEE*

Abstract—Anchor graph-based clustering has demonstrated strong potential for uncovering complex latent structures in large-scale scenarios. Nevertheless, existing approaches face two critical limitations: first, most fail to fully exploit the deep structural relationships among samples, resulting in graphs that inadequately capture the intrinsic data topology; second, the conventional two-stage paradigm that separates spectral embedding from label assignment introduces relaxation errors and redundant computations, degrading clustering performance and increasing computational overhead. To address these challenges, we propose an Efficient Structure-Aware Discrete Clustering via Multi-Order Anchor Graphs (ESADC). ESADC adaptively fuses multi-order anchor graphs to model complementary approximations of the underlying continuous manifold, while employing a single-stage, structure-aware framework that jointly learns spectral embeddings and discrete cluster labels, thereby enhancing both clustering effectiveness and computational efficiency. Furthermore, a fast coordinate descent-based optimization algorithm is developed for the discrete ESADC model to accelerate convergence. Extensive experiments on both regular and large-scale real-world datasets demonstrate that ESADC consistently outperforms state-of-the-art methods, highlighting its efficiency and strong structure-aware capability.

Index Terms—Discrete clustering, anchor graphs, large-scale datasets, and multi-order structure.

I. INTRODUCTION

CLUSTERING, as a fundamental paradigm in unsupervised learning, seeks to reveal latent data structures by

organizing unlabeled samples into semantically coherent clusters through similarity maximization [1], [2], [3], [4]. It enables effective data partitioning without prior labeling and plays a pivotal role in domains such as computer vision [5], bioinformatics [6], and data mining [7]. Representative methods such as K -means [8] and non-negative matrix factorization (NMF) [9] are efficient and interpretable, yet can be limited in modeling complex nonlinear structures and are susceptible to high-dimensional noise. In contrast, graph-based clustering methods [10], [11], [12] provide a more expressive framework for modeling data dependencies, thereby facilitating more robust and insightful knowledge discovery. Nevertheless, conventional graph-based methods often suffer from high computational and memory costs. This substantial complexity severely limits their efficiency, making them impractical for large-scale datasets commonly encountered in real-world applications.

To alleviate this issue, recent studies have introduced anchor-based graph clustering [13], [14], [15], which approximates the full similarity graph by introducing a small set of representative anchor points. By constructing a compact sample-anchor bipartite graph, these methods largely reduce the time and memory costs of graph construction while retaining essential manifold information. Owing to these advantages, anchor-based clustering has gained wide adoption in large-scale applications. Representative efforts include anchor graph regularization [16], self-supervised anchor graph learning [17], bipartite propagation-based formulations [13], and anchor-based ensemble clustering [18]. These advances show that anchor-based graph clustering provides an efficient, scalable, and high-performance solution for large-scale scenarios. Despite these benefits, existing anchor-based graph models still face two critical challenges: (i) they insufficiently capture complex structural relationships, leading to an incomplete description of the intrinsic data topology; and (ii) their two-stage design, which separates spectral embedding from label assignment, inevitably incurs relaxation errors and redundant computations, constraining both accuracy and efficiency.

To address the first challenge of insufficient topology preservation, various extended graph modeling strategies have been proposed, including hypergraphs [19], [20], multi-view graphs [21], [22], and higher-order graphs [23], [24], each providing distinct improvements over conventional pairwise graphs. Hypergraph-based methods enhance the expressive power of graph structures by allowing hyperedges to connect multiple vertices simultaneously. For example, Li et al. [19]

Received 24 September 2025; revised 2 April 2026; accepted 1 June 2026. Date of publication 8 June 2026; date of current version 8 August 2026. This work was supported in part by the MOE (Ministry of Education in China) Liberal Arts and Social Sciences Foundation under Grant 23YJCZH336, in part by the National Natural Science Foundation of China under Grant 62403372, in part by the Natural Science Foundation of Sichuan Province under Grant 2025ZNSFSC1496, in part by the National Postdoctoral Innovative Talents Support Program under Grant BX20230283, in part by the “Sanqin Bochuang” Talent Support Program of Shaanxi Province under Grant 2024SQBC005, in part by China Postdoctoral Science Foundation under Grant 2023M742791, and in part by the Fundamental Research Funds for the Central Universities, Zhongnan University of Economics and Law under Grant 2722025BJ120. Recommended for acceptance by Y. Yuan. (*Corresponding author: Yu Zhou.*)

Ben Yang, Xuetao Zhang, and Badong Chen are with the State Key Laboratory of Human-Machine Hybrid Augmented Intelligence, Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University, Xi'an 710049, China (e-mail: benyang@xjtu.edu.cn; xuetaozhang@xjtu.edu.cn; bdchen@mail.xjtu.edu.cn).

Yu Zhou and Haoxin Wu are with the School of Information Engineering, Zhongnan University of Economics and Law, Wuhan 430073, China (e-mail: zhouyuei@gmail.com; 1782795967@qq.com).

Feiping Nie is with the School of Computer Science and School of Artificial Intelligence, Optics and Electronics (iOPEN), Northwestern Polytechnical University, Xi'an 710072, China (e-mail: feipingnie@gmail.com).

Digital Object Identifier 10.1109/TKDE.2026.3700798

proposed an inhomogeneous hypergraph partitioning approach that enhances flexibility by assigning different costs to different cuts. Similarly, the k-HyperEdge medoids method [20] selects hyperedges from the clustering view and refines them from the sample view, yielding effective and robust ensemble clustering. Although hypergraphs offer these advantages, most approaches employ binary or uniform edge weights, which limit their ability to model fine-grained local variations and consequently reduce clustering performance. Multi-view methods aim to integrate complementary information from different feature spaces to obtain more comprehensive data representations. For instance, Li et al. [21] present a multi-view graph-based hierarchical representation learning to mine organized money laundering groups, achieving superior performance on public transaction datasets. Liu et al. [22] proposed a multi-view multi-label classification framework that uses label-driven contrastive learning and quality-aware view weighting to maintain intra-view structure and enforce cross-view consistency. While these methods offer advantages in robustness and cross-view representation, the heterogeneity among views introduces challenges for alignment and fusion, resulting in suboptimal consensus. Recent studies [25], [26], [27], [28] have explored collaborative structure learning [25], multi-granularity ensemble [26], and adaptive similarity fusion [27] to enhance clustering robustness and representation capability. These works further motivate the importance of exploiting complementary structural information from different perspectives. In contrast, high-order graph-based methods can approximate continuous manifolds through multi-step connectivity, providing a more natural means of preserving both global and local data structures. For instance, Tang et al. [23] proposed LINE, a network embedding method for large-scale information networks that preserves undirected and directed structures, enabling scalable and effective analysis across various datasets. Zhao et al. [24] constructs a high-order bipartite graph proximity matrix and learns a low-rank tensor representation to obtain a structured proximity matrix, yielding improved clustering performance. These methods demonstrate superior performance in manifold smoothness approximation and modeling of long-range relationships. Nevertheless, relying solely on single-order connectivity is insufficient to capture the coexisting linear and nonlinear manifold characteristics inherent in real-world data, which may lead to incomplete topological representations and introduce noise. Therefore, it is imperative to investigate and develop new higher-order graph-based methods for effectively capturing multi-scale structural features of complex data.

The second challenge stems from the two-stage paradigm. Since deriving discrete cluster indicators is an NP-hard problem, traditional methods [21], [29], [30] typically perform eigen-decomposition of the graph Laplacian to obtain a continuous embedding, followed by K -means or spectral rotation [31] for discretization. This two-step separation often causes the loss of essential structural information and incurs cubic-time complexity, thereby limiting its applicability to large-scale datasets. To address this, recent studies [32], [33], [34] have sought to unify embedding and clustering into a single framework. Existing one-step discretization strategies can be classified into

two categories. The first category integrates the embedding stage with clustering; for instance, UOMvSC [35] unifies spectral embedding and K -means into a single stage, yet its reliance on full-sample graphs severely limits scalability. The second introduces auxiliary variables to achieve one-step discretization; for example, E2OMVC [36] incorporates label learning into latent information fusion, while OMVCDR [37] imposes a regularization term to derive a consensus discrete partition matrix. However, these inevitably increase computational complexity by introducing additional variables. Hence, developing an efficient one-step discrete framework that preserves structural information while avoiding costly eigen-decomposition remains an essential task.

Drawing on the above analyses, we propose Efficient Structure-Aware Discrete Clustering via Multi-Order Anchor Graphs (ESADC) to jointly address the limitations of insufficient structural modeling and the inefficiency of the two-stage relaxation. Specifically, ESADC first generates candidate anchors and constructs multi-order adjacency matrices through similarity diffusion, where each order represents the data manifold at a different geometric scale: lower orders retain local fidelity, whereas higher orders encode global consistency. These multi-order graphs are then adaptively fused into a consensus graph by learning order-specific weights, which mitigates structural bias and jointly preserves local neighborhoods and global topology, yielding a robust and discriminative manifold representation. On top of this representation, ESADC introduces an efficient one-step clustering strategy that directly derives discrete cluster assignments from the consensus graph. Unlike prior one-step methods that suffer from local inconsistencies or expensive repeated decompositions, our approach enforces global consistency while maintaining near-linear complexity. To satisfy discrete constraints and guarantee convergence, we further design a decomposed alternating minimization scheme, ensuring scalability for large-scale applications. As depicted in Fig. 1, ESADC unifies multi-order anchor graph fusion and discrete clustering within a single architecture. Moreover, Table I presents the definitions of important mathematical notations. The main breakthroughs of this paper are:

- We propose ESADC, a unified one-stage clustering framework that constructs anchor-based multi-order adjacency matrices and adaptively fuses them, thereby capturing both fine-grained local connectivity and global manifold structures.
- By performing discrete spectral analysis directly on the fused consensus graph, ESADC eliminates the need for two-step relaxation and costly eigen-decomposition, thereby retaining critical structural information and enhancing clustering precision.
- We devise a decomposed alternating minimization strategy to efficiently solve the discrete optimization problem, ensuring both scalability and rapid convergence.
- Comprehensive experiments on multiple real-world benchmark datasets demonstrate that ESADC consistently outperforms state-of-the-art methods in terms of both effectiveness and efficiency.

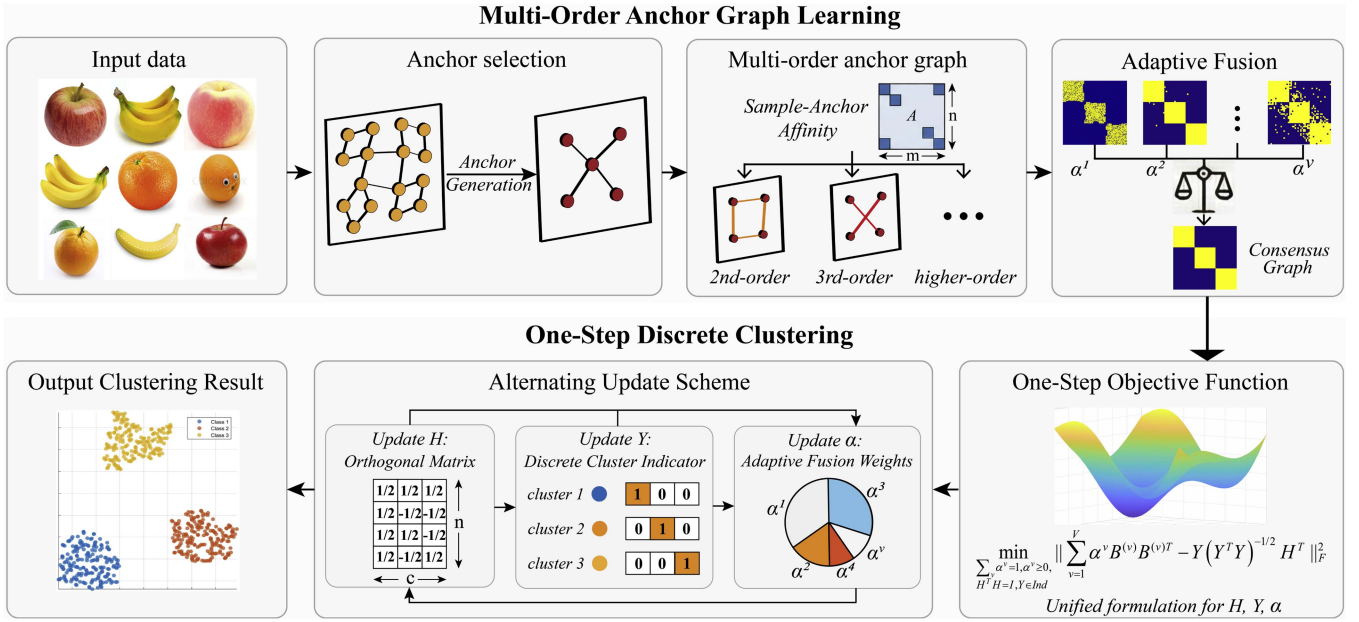


Fig. 1. The framework of the proposed ESADC.

TABLE I
IMPORTANT SYMBOLS AND ABBREVIATIONS

Notation	Meaning
X	Original samples
V	Vertices of graph
E	Edges of graph
S	Similarity matrix
D	Degree of matrix
L	Laplacian matrix of graph
Y	Discrete cluster indicator matrix
G	Graph representation matrix
G^T	The transpose of G
$Tr(G)$	The trace of G
$\ G\ _F$	The Frobenius norm of G
G^n	The n th column of G
H	Orthogonal spectral embedding
B	High-order anchor graph matrix
A	The affinity of a graph
Z	The anchor set of samples
$\mathbf{1}$	A column vector with all the elements as one
I	The identity matrix
α	The weight of high-order graph
v	The number of orders of the graph
n	The number of samples
d	The feature dimension
c	The number of clusters

II. PRELIMINARIES

A. Graph Cutting

Let a weighted graph $G = (V, E)$ have node features $X \in \mathbb{R}^{n \times d}$ and similarity matrix $S \in \mathbb{R}^{n \times n}$. Defines the diagonal degree matrix D by $D_{ii} = \sum_{j=1}^n S_{ij}$, and the (unnormalized) Laplacian $L = D - S$. Graph cutting seeks a partition $\{X_1, X_2, \dots, X_c\}$ of the samples minimizing inter-cluster cuts:

$$cut(X_l, \bar{X}_l) = \sum_{i \in X_l} \sum_{j \notin X_l} S_{ij} \quad (1)$$

Ratio cut and normalized cut are respectively normalized by cluster size or volume:

$$Rcut(X_1, \dots, X_c) = \min \sum_{l=1}^c \frac{cut(X_l, \bar{X}_l)}{|X_l|} \quad (2)$$

$$Ncut(X_1, \dots, X_c) = \min \sum_{l=1}^c \frac{cut(X_l, \bar{X}_l)}{vol(X_l)} \quad (3)$$

Using a binary indicator matrix $Y \in \{0, 1\}^{n \times c}$, Rcut objective can be rewritten in trace form:

$$\min_{Y \in Ind} Tr \left[(Y^T Y)^{-1/2} Y^T L Y (Y^T Y)^{-1/2} \right] \quad (4)$$

Because this discrete problem is NP-hard, it can be relaxed by defining $Z = Y(Y^T Y)^{-1/2}$, with $Z^T Z = I$, yields

$$\min_{Z^T Z = I} Tr(Z^T L Z) \quad (5)$$

whose solution is the c eigenvectors of L with smallest eigenvalues. Ncut follows the same steps using the normalized Laplacian $\tilde{L} = D^{-1/2} L D^{-1/2}$. The resulting spectral embedding is then discretized by K -means.

B. High-Order Graph

A high-order graph captures multi-hop relationships between nodes by repeatedly multiplying the original adjacency matrix. Specifically, the affinity of a graph is defined by A , and the second-order graph can be expressed as $A_2 = A A^T$. Due to the symmetry of A , it can be also expressed as $A_2 = A A$. Hence, the k th order graph can be recursively constructed as

$$A_k = \begin{cases} A_{k-1} A_1, & k > 1, \\ A, & k = 1. \end{cases} \quad (6)$$

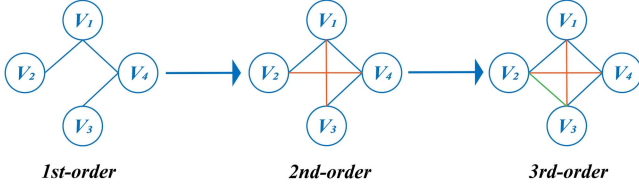


Fig. 2. The property of a high-order graph.

Propagating affinities along multi-hop paths converts indirect relationships into direct links in the k -order graph. As depicted in Fig. 2, vertices v_1 and v_3 are connected only through a two-step path via v_4 in the original graph, but these two nodes can acquire a direct edge in the second-order graph. Furthermore, when k is raised to three, vertices such as v_2 and v_3 , which originally lay three hops apart, become immediate neighbors. It shows the high-order characteristic of converting the neighbors of neighbors to neighbors. Nevertheless, conventional high-order graph methods are limited to homogeneous $n \times n$ affinity matrices, while real-world heterogeneous data often appears as $n \times m$ with variable width. Thus, developing a high-order graph representation tailored to such uncertain structures is essential.

C. Anchor Graph Learning

Anchor graph learning offers a scalable alternative to constructing a full pairwise similarity graph by introducing a compact set of representative anchors $\{z_j\}_{j=1}^m$ with $m \ll n$. Instead of computing all pairwise similarities, it builds a sparse anchor-sample affinity matrix $\tilde{A}_{ij} \in \mathbb{R}^{n \times m}$, where each row $\tilde{A}_i$ encodes how sample x_i is connected to its nearest anchors. Specifically, $\tilde{A}_i$ is obtained by solving a locally constrained reconstruction problem:

$$\min_{\tilde{A}_i} \sum_{j=1}^m \left(\|x_i - z_j\|_2^2 \tilde{A}_{ij} + \xi \tilde{A}_{ij}^2 \right) \quad (7)$$

where $\tilde{A}_{ij}$ satisfies $\mathbf{1}^\top \tilde{A}_i = 1$, $\tilde{A}_i \geq 0$, and ξ is an adaptive regularization parameter determined by the distance to the $(k+1)$ -th nearest anchor distance. The closed-form solution produces a row-sparse $\tilde{A}$ in which only the k nearest anchors have nonzero weights. This anchor-based representation effectively reduces the complexity of graph construction from $O(n^2)$ to $O(nm)$. Once $\tilde{A}$ is constructed, an approximate similarity graph can be obtained by $\tilde{S} = \tilde{A}\tilde{A}^\top$. Finally, this resulting approximate similarity graph is used as a compact and efficient representation, making anchor graph learning particularly suitable for large-scale and high-dimensional datasets.

III. METHODOLOGY

This paper introduces ESADC, which first builds multi-order sample-to-anchor graphs with immediate diffusion. Then, it adaptively fuses them into one consensus graph and directly solves discrete spectral clustering on the fused similarity matrix in a single step. Finally, a decomposed alternating minimization

scheme ensures efficient convergence without requiring eigen-decomposition or additional parameters. Detailed formulation and optimization follow.

A. High-Order Anchor Graph Generation

The notion of order characterizes the depth of affinity propagation through the anchor space. Starting from a sample-anchor affinity matrix, different orders correspond to aggregating similarities along anchor-hop paths of increasing length, thereby capturing complementary structural information from local neighborhoods to global manifold organization. To operationalize this idea, it first constructs a compact anchor-based affinity representation and then derives a family of order-specific graphs in a unified manner. Specifically, the high-order anchor graph framework begins by organizing the feature matrix $\mathbf{X} = [x_1, x_2, \dots, x_n]^\top \in \mathbb{R}^{n \times d}$, where each row vector x_i represents the d -dimensional features of sample i . It then applies the K -means clustering algorithm on $\mathbf{X}$ to partition the data into m clusters, and uses the resulting cluster centroids as anchors. To reduce the sensitivity of anchor selection to initialization, we run K -means 10 times with different random initializations, and the final anchors are obtained by averaging the anchor sets produced across these runs. This strategy improves the stability of anchor generation while introducing only a minor additional cost. For each sample x_i , reconstruction weights are computed over its k nearest anchors to form a sparse sample-anchor affinity matrix. The resulting anchor set is denoted by

$$\mathbf{Z} = (z_1, z_2, \dots, z_m), \quad z_j \in \mathbb{R}^d \quad (8)$$

where each anchor corresponds to a cluster centroid and is expected to represent the underlying data distribution. Based on these anchors, we construct an unnormalized sample-anchor affinity matrix $\tilde{A} \in \mathbb{R}^{n \times m}$, whose entries are defined via (7). To balance the influence of samples and anchors, $\tilde{A}$ may be row- or column-normalized, producing the affinity matrix $\mathbf{A}$. Using $\mathbf{A}$ as the first-order sample-anchor representation, it defines

$$\mathbf{B}^{(1)} = \mathbf{A} \quad (9)$$

where $\mathbf{B}^{(1)}$ describes the direct associations between samples and anchors. To further capture higher-order and indirect structural dependencies, it can define the v -th order graph by

$$\mathbf{B}^{(v)} = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{v-1}, \quad v = 2, 3, \dots, V \quad (10)$$

where $\mathbf{A}^\top \mathbf{A} \in \mathbb{R}^{m \times m}$ is the anchor self-similarity matrix. In this way, higher-order representations are obtained by propagating structural information through the anchor space. For instance, $\mathbf{B}^{(2)}$ and $\mathbf{B}^{(3)}$ correspond to one-step and two-step anchor propagation, respectively. In general, $\mathbf{B}^{(v)}$ captures sample-anchor relationships after v -step structural propagation and serves as the v -th order anchor graph.

B. Multi-Order Anchor Graph Fusion

The different-order anchor graph representations provide complementary views of the underlying manifold that a single order alone cannot fully represent. Lower-order graphs mainly preserve local neighborhood structure, whereas higher-order

graphs progressively reveal broader and more global dependencies. To exploit these complementary properties, ESADC adaptively fuses the order-specific anchor graph representations into a single consensus graph, so as to better reflect the intrinsic geometry of the data.

Specifically, for each order v , the corresponding sample-sample structural affinity is induced by $\mathbf{B}^{(v)} (\mathbf{B}^{(v)})^\top$, where the corresponding sample-anchor representation $\mathbf{B}^{(v)} \in \mathbb{R}^{n \times m}$. This matrix encodes a distinct structural pattern corresponding to the v -th propagation scale in the anchor space. To formulate the contributions of different orders, we introduce non-negative fusion weights:

$$\boldsymbol{\alpha} = [\alpha^{(1)}, \alpha^{(2)}, \dots, \alpha^{(V)}]^\top, \quad \sum_{v=1}^V \alpha^{(v)} = 1 \quad (11)$$

Hence, the final consensus adjacency matrix is obtained by a weighted sum of the order-specific interaction matrices:

$$\hat{\mathbf{B}} = \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} (\mathbf{B}^{(v)})^\top \quad (12)$$

where $\alpha^{(v)}$ is initially set proportional to a measure of graph quality. Through this adaptive fusion, it integrates complementary local-to-global structural information into a unified representation, which is more robust and informative for subsequent clustering.

C. One-Step Discrete Clustering

Traditional graph clustering [21], [29] typically relaxes discrete assignments via Laplacian eigen-analysis and then recovers cluster labels via a separate discretization. Such a two-stage pipeline may introduce structural information loss and substantial computational cost. To address this issue, ESADC unifies graph representation and discrete assignment within a single structure-aware framework.

Specifically, classical graph cutting objectives such as Rcut and Ncut can be equivalently reformulated from the trace minimization in (5) as the following graph reconstruction problem under the Frobenius norm:

$$\min_{\mathbf{Y} \in \text{Ind}} \|\mathbf{S} - \mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-1} \mathbf{Y}^\top\|_F^2 \quad (13)$$

where $\mathbf{Y}$ is binary cluster indicator matrix and $\mathbf{S}$ is similarity matrix. This formulation lays the foundation for replacing the original similarity matrix $\mathbf{S}$ with a multi-order anchor consensus graph representation.

Based on the order-specific representations $\mathbf{B}^{(v)}$, ESADC constructs a consensus graph by adaptively combining $\mathbf{B}^{(v)} \mathbf{B}^{(v)\top}$ with nonnegative fusion weights $\alpha^{(v)}$. The corresponding one-step clustering objective is written as

$$\min_{\substack{\boldsymbol{\alpha}^\top \mathbf{1} = 1, \alpha \geq 0, \\ \mathbf{Y} \in \text{Ind}}} \left\| \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top} - \mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-1} \mathbf{Y}^\top \right\|_F^2 \quad (14)$$

where the second term is a normalized projection matrix that characterizes the ideal clustering assignment structure.

However, directly optimizing $\mathbf{Y}$ in this form may suffer from row-norm instability. To preserve the one-hot sparsity of $\mathbf{Y}$ while improving numerical stability, we use its normalized form $\mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-1/2}$ and further introduce an orthogonal rotation matrix $\mathbf{H} \in \mathbb{R}^{n \times c}$ satisfying $\mathbf{H}^\top \mathbf{H} = \mathbf{I}$. The role of $\mathbf{H}$ is to relax the strict one-hot constraint by allowing an orthogonal transformation in the c -dimensional embedding space while preserving the normalized geometry of the clustering structure. In this way, the feasible space is enlarged without destroying the relative structural relationships, thereby improving optimization stability. Accordingly, the final one-step objective is formulated as

$$\min_{\substack{\boldsymbol{\alpha}^\top \mathbf{1} = 1, \alpha \geq 0, \\ \mathbf{H}^\top \mathbf{H} = \mathbf{I}, \mathbf{Y} \in \text{Ind}}} \left\| \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top} - \mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-1/2} \mathbf{H}^\top \right\|_F^2 \quad (15)$$

This formulation directly aligns the fused consensus graph with the rotated normalized indicator structure, thereby tightly coupling graph learning and discrete assignment, reducing the mismatch between structural representation and final labels, and avoiding the conventional two-stage relaxation pipeline. Moreover, this joint optimization can be solved by alternating minimization. Each update step involves only low-complexity operations such as matrix multiplications, simplex projections, and local SVDs, avoiding eigen-decomposition and ensuring linear scalability for the sample size.

IV. OPTIMIZATION AND ANALYSIS

Equation (15) defines the objective function of ESADC. The binary indicator matrix $\mathbf{Y}$ is initialized using an anchor-based spectral embedding derived from the first-order anchor graph, followed by K -means clustering. The resulting labels are converted into a binary assignment matrix. Next, we build the high-order anchor graph similarity matrix $\mathbf{B}^{(v)}$ as (10) and initialize $\alpha^{(v)}$, $\mathbf{H}$ (with $\mathbf{H}^\top \mathbf{H} = \mathbf{I}$) via SVD on $\alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top}$. To tackle the three-variable optimization, we employ an alternating minimization strategy, updating $\mathbf{H}$, $\mathbf{Y}$, and $\boldsymbol{\alpha}$ in turn to decompose the original problem into three tractable sub-problems. By iteratively minimizing until the objective converges, it ultimately derives the sample clusters from the $\mathbf{Y}$. The entire procedure appears as Algorithm 1.

A. Update $\mathbf{H}$

When solving subproblem of $\mathbf{H}$ (with $\boldsymbol{\alpha}$ and $\mathbf{Y}$ fixed), Algorithm 1 can describe the updating process. Specifically, the objective function (15) can be rewritten to:

$$\min_{\mathbf{H}^\top \mathbf{H} = \mathbf{I}} \|\mathbf{S}(\boldsymbol{\alpha}) - \tilde{\mathbf{Y}} \mathbf{H}^\top\|_F^2 \quad (16)$$

where $\mathbf{S}(\boldsymbol{\alpha}) = \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top}$, $\tilde{\mathbf{Y}} = \mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-1/2}$ and $\|\cdot\|_F$ denotes the Frobenius norm, with the constraint $\mathbf{H}^\top \mathbf{H} = \mathbf{I}$ enforcing that $\mathbf{H}$ is an orthogonal matrix. Expanding (16) and using $\tilde{\mathbf{Y}}^\top \tilde{\mathbf{Y}} = \mathbf{I}$, this is equivalent to

$$\max_{\mathbf{H}^\top \mathbf{H} = \mathbf{I}} \text{Tr}(\mathbf{H} \mathbf{M}), \quad \mathbf{M} = \tilde{\mathbf{Y}}^\top \mathbf{S} \quad (17)$$

Algorithm 1: Algorithm of ESADC.

Input: Feature $\mathbf{X} \in \mathbb{R}^{n \times d}$, anchors m , orders V , clusters c , number of nearest anchors k , bandwidth σ
Output: Discrete indicator matrix $\mathbf{Y} \in \{0, 1\}^{n \times c}$

- 1: **Anchor Selection:**
 Run K -means on $\mathbf{X}$ for 10 times under different random initializations, and obtain the final anchors by averaging the anchors generated from these runs.
 $\mathbf{Z} = \{z_1, \dots, z_m\}$.
- 2: **High-order Graph Construction:**
 Compute unnormalized affinities $\tilde{A}_{ij} \quad \triangleright (7)$
 Normalize rows or columns of $\tilde{A}$ to get $\mathbf{A} \in \mathbb{R}^{n \times m}$.
for $v = 1, \dots, V$ **do**
 $\mathbf{B}^{(1)} \leftarrow \mathbf{A} \quad \triangleright (9)$
 $\mathbf{B}^{(v)} \leftarrow \mathbf{A} (\mathbf{A}^\top \mathbf{A})^{v-1} \quad (v \geq 2) \quad \triangleright (10)$
end for
- 3: **Initialization:**
 Initialize fusion weights $\alpha = [\alpha^{(1)}, \dots, \alpha^{(V)}]^\top$
 Initialize binary labels $\mathbf{Y}$ and orthogonal matrix $\mathbf{H} \in \mathbb{R}^{n \times c}$.
- 4: **Fuse Multi-order Anchor Graphs:**
for $v = 1, \dots, V$ **do**
 $\hat{\mathbf{B}} \leftarrow \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top} \quad \triangleright (12)$
end for
- 5: **One-step Discrete Clustering:**
while not converged **do**
 Solve

$$\min_{\mathbf{Y}, \alpha, \mathbf{H}} \left\| \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top} - \mathbf{Y} (\mathbf{Y}^\top \mathbf{Y})^{-\frac{1}{2}} \mathbf{H}^\top \right\|_F^2 \quad \triangleright (15)$$

 (a) **Update** $\mathbf{H}$ via orthogonal Procrustes $\triangleright$ **Algorithm 2**
 (b) **Update** $\mathbf{Y}$ via row-wise minimization $\triangleright$ **Algorithm 3**
 (c) **Update** α via simplex projection $\triangleright$ **Algorithm 4**
end while
- 6: **Return** final clustering labels from $\mathbf{Y}$.

It is distinct that the maximization $\max_{\mathbf{H}^\top \mathbf{H} = \mathbf{I}} \text{Tr}(\mathbf{H} \mathbf{M})$ is the classical orthogonal Procrustes problem. The closed-form solution via the singular value decomposition (SVD) can be:

$$\mathbf{M} = \mathbf{U} \mathbf{\Sigma} \mathbf{V}^\top, \quad \mathbf{H} = \mathbf{V} \mathbf{U}^\top \quad (18)$$

where $\mathbf{U} \in \mathbb{R}^{c \times c}$ and $\mathbf{V} \in \mathbb{R}^{n \times c}$ are orthogonal matrices, and $\mathbf{\Sigma} \in \mathbb{R}^{c \times c}$ is diagonal matrix of singular values. To avoid forming a large $c \times n$ matrix $\mathbf{M}$ and computing its full SVD, it can define $\mathcal{N} = \mathbf{S}(\alpha) \tilde{\mathbf{Y}} \in \mathbb{R}^{n \times c}$, which shares the same nonzero singular values and singular vectors with $\mathbf{M}$. Since $\mathbf{S}(\alpha)$ is symmetric, it can be inferred that:

$$\mathcal{N}^\top \mathcal{N} = \tilde{\mathbf{Y}}^\top \mathbf{S}(\alpha)^2 \tilde{\mathbf{Y}} = \mathbf{M} \mathbf{M}^\top \quad (19)$$

Since each $\mathbf{B}^{(v)}$ is an $n \times m$ low-rank factor ($m \ll n$), it utilizes a compact SVD of $\mathcal{N} = \mathbf{V} \mathbf{\Sigma} \mathbf{U}^\top$ directly yields $\mathbf{H} = \mathbf{V} \mathbf{U}^\top$ at a cost of $O(nc^2)$. This compact decomposition efficiently captures the most informative subspace of $\mathcal{N}$, avoiding

Algorithm 2: Updating $\mathbf{H}$.

Input: Discrete indicator matrix $\tilde{\mathbf{Y}} \in \mathbb{R}^{n \times c}$, The fused high-order anchor graph similarity matrix $\mathbf{S}(\alpha) = \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top} \in \mathbb{R}^{n \times n}$
Output: Orthogonal matrix $\mathbf{H} \in \mathbb{R}^{n \times c}$ satisfying $\mathbf{H}^\top \mathbf{H} = \mathbf{I}$

- 1: Form normalized indicators:
 $\tilde{\mathbf{Y}} \leftarrow \mathbf{Y} (\mathbf{Y}^\top \mathbf{Y})^{-1/2} \in \mathbb{R}^{n \times c}$
- 2: Build the thin matrix:
 $\mathcal{N} \leftarrow \mathbf{S}(\alpha) \tilde{\mathbf{Y}} \in \mathbb{R}^{n \times c}$
- 3: Compute compact SVD of $\mathcal{N}$:
 $[\mathbf{V}, \mathbf{\Sigma}, \mathbf{U}^\top] \leftarrow \text{SVD}(\mathcal{N})$ (columns of $\mathbf{V} \in \mathbb{R}^{n \times c}$, $\mathbf{U} \in \mathbb{R}^{c \times c}$)
- 4: Construct optimal rotation:
 $\mathbf{H} \leftarrow \mathbf{V} \mathbf{U}^\top \quad \triangleright \text{Eq. (18)}$
- 5: **return** $\mathbf{H}$

unnecessary computations on redundant directions. The entire procedure is linear in n and thus scalable to large-scale datasets, improving the overall efficiency of ESADC.

B. Update $\mathbf{Y}$

Algorithm 3 demonstrates the process of updating $\mathbf{Y}$. With α and $\mathbf{H}$ fixed, $\mathbf{Y} \in \{0, 1\}^{n \times c}$ is equivalently formulated as:

$$\max_{\mathbf{Y} \in \text{Ind}} \text{Tr}(\mathbf{H} ((\mathbf{Y} (\mathbf{Y}^\top \mathbf{Y})^{-1/2}))^\top \mathbf{S}) \quad (20)$$

Setting $\mathcal{G} = \mathbf{S} \mathbf{H} \in \mathbb{R}^{n \times c}$ and using low-rank structure to avoid the calculation of an $n \times n$ matrix. Equation (20) becomes:

$$\text{Tr}(\mathbf{H} (\mathbf{Y} (\mathbf{Y}^\top \mathbf{Y})^{-1/2})^\top \mathbf{S}) = \text{Tr}((\mathbf{Y} (\mathbf{Y}^\top \mathbf{Y})^{-1/2})^\top \mathcal{G}) \quad (21)$$

With $\mathbf{Y}^\top \mathbf{Y}$ is a diagonal matrix, and the (n, n) th component of $\mathbf{Y}^\top \mathbf{Y}$ is equal to $\mathbf{Y}_{.n}^\top \mathbf{Y}_{.n}$. Hence, (21) can be

$$\max_{\mathbf{Y} \in \text{Ind}} \sum_{n=1}^c \frac{(\mathbf{Y}_{.n})^\top \mathcal{G}_{.n}}{\sqrt{\mathbf{Y}_{.n}^\top \mathbf{Y}_{.n}}} \quad (22)$$

with $\mathbf{Y}_{.n}$ and $\mathcal{G}_{.n}$ are the n th columns of $\mathbf{Y}$ and $\mathcal{G}$, c is the number of clusters. Because the column norms in the denominator couple all rows, a global update is intractable. Hence, row-by-row coordinate descent can be used. For each sample i , it tests all c possible clusters and selects the one that maximizes the objective, e.g., i th row is associated with c clusters. Each cluster is the j th cluster where $j \in \{1, 2, \dots, c\}$:

$$\mathbf{Y}^j = \begin{bmatrix} \dots & \mathbf{Y}_{ij} \\ \vdots & \downarrow \\ 0 & 0 & \dots & 0 & 1 & \dots & 0 \\ \vdots & \\ \dots & \end{bmatrix} \quad (23)$$

It denote by $\mathbf{Y}^l$ the matrix obtained by setting row i of $\mathbf{Y}$ to one-hot vector at position l . Then (22) reduces to

$$\max_{l \in \{1, \dots, c\}} \sum_{n=1}^c \frac{(\mathbf{Y}_{.n}^l)^\top \mathcal{G}_{.n}}{\sqrt{(\mathbf{Y}_{.n}^l)^\top \mathbf{Y}_{.n}^l}} \quad (24)$$

Algorithm 3: Updating $\mathbf{Y}$.

Input: Fused similarity matrix $\mathbf{S}(\alpha) \in \mathbb{R}^{n \times n}$, Orthogonal matrix $\mathbf{H} \in \mathbb{R}^{n \times c}$, Initial indicator $\mathbf{Y} \in \{0, 1\}^{n \times c}$ (one-hot rows)

Output: Updated indicator $\mathbf{Y} \in \{0, 1\}^{n \times c}$

- 1: Precompute affinity: $\mathcal{G} \leftarrow \mathbf{S}(\alpha)\mathbf{H} \in \mathbb{R}^{n \times c}$
- 2: Compute $\Delta \leftarrow \mathbf{Y}^\top \mathbf{Y}$ and its Cholesky factor $\mathbf{V}$ (so $\mathbf{V}^\top \mathbf{V} = \Delta$)
- 3: **repeat**
- 4: changed $\leftarrow$ false
- 5: **for** $i = 1$ **to** n **do**
- 6: Let $p = \arg\{j \mid \mathbf{Y}_{i,j} = 1\}$
- 7: **for** $l = 1$ **to** c **do**
- 8: Computing $\phi(l)$ $\triangleright$ Eq. (26)
- 9: **end for**
- 10: $j \leftarrow \arg \max_{l=1, \dots, c} \phi(l)$
- 11: **if** $j \neq p$ **then**
- 12: $\mathbf{Y}_{i,p} \leftarrow 0$, $\mathbf{Y}_{i,j} \leftarrow 1$ $\triangleright$ Eq. (27)
- 13: changed $\leftarrow$ true
- 14: **end if**
- 15: **end for**
- 16: **until** not changed
- 17: **return** $\mathbf{Y}$

where $\mathbf{Y}_{.n}^l$ is the n -th column of the candidate matrix $\mathbf{Y}^l$. Introducing $\mathbf{Y}^0$ (identical to $\mathbf{Y}^l$ but with row i zeroed) and substituting into (24) yields the equivalent formulation:

$$\max_{l \in (1, \dots, c)} \phi(l) = \max_{l \in (1, \dots, c)} \left(\frac{(\mathbf{Y}_{.l}^l)^\top \mathcal{G}_{.n}}{\sqrt{(\mathbf{Y}_{.l}^l)^\top \mathbf{Y}_{.l}^l}} - \frac{(\mathbf{Y}_{.l}^0)^\top \mathcal{G}_{.n}}{\sqrt{(\mathbf{Y}_{.l}^0)^\top \mathbf{Y}_{.l}^0}} \right) \quad (25)$$

For each i , the gain $\phi(l)$ measures the change in objective when its label moves from cluster p to l . Using Cholesky factor $\mathbf{V}$ of $\Delta = \mathbf{Y}^\top \mathbf{Y}$, $\phi(l)$ admits two closed-form cases:

$$\phi(l) = \begin{cases} \frac{\mathbf{Y}_{.l}^\top \mathbf{V}^\top \mathbf{V} \mathbf{Y}_{.l} - \mathbf{Y}_{.l}^\top \mathbf{V}^\top \mathbf{V} \mathbf{Y}_{.l-2V_{.i}^\top \mathbf{V} \mathbf{Y}_{.l} + V_{.i}^\top \mathbf{V}_{.i}}}{\mathbf{Y}_{.l}^\top \mathbf{Y}_{.l-1}}, & l = p \\ \frac{\mathbf{Y}_{.l}^\top \mathbf{V}^\top \mathbf{V} \mathbf{Y}_{.l+2V_{.i}^\top \mathbf{V} \mathbf{Y}_{.l} + V_{.i}^\top \mathbf{V}_{.i}}}{\mathbf{Y}_{.l}^\top \mathbf{Y}_{.l+1}} - \frac{\mathbf{Y}_{.l}^\top \mathbf{V}^\top \mathbf{V} \mathbf{Y}_{.l}}{\mathbf{Y}_{.l}^\top \mathbf{Y}_{.l}}, & l \neq p \end{cases} \quad (26)$$

Once all $\phi(l)$ are obtained, it updates the i -th row of $\mathbf{Y}$ by

$$\mathbf{Y}_{ij} = \begin{cases} 1 & , \quad j = \arg \max_{l \in 1, \dots, c} \phi(l) \\ 0 & , \quad \text{otherwise} \end{cases} \quad (27)$$

and repeat over all rows to obtain the converged $\mathbf{Y}$.

C. Update α

With $\mathbf{H}$ and $\mathbf{Y}$ fixed, updating α can be transformed into

$$\min_{\alpha^\top \mathbf{1} = 1, \alpha \geq 0} \|\mathbf{S}(\alpha) - \tilde{\mathbf{Y}} \mathbf{H}^\top\|_F^2 \quad (28)$$

where $\mathbf{S}(\alpha) = \sum_{v=1}^V \alpha^{(v)} \mathbf{K}^{(v)}$, $\mathbf{K}^{(v)} = \mathbf{B}^{(v)} \mathbf{B}^{(v)\top}$. It demonstrates that optimizing α reduces to a convex quadratic

Algorithm 4: Update α .

Input: Matrix $\mathcal{A}$ and row vector b .

Output: Column vector α .

- 1: **Initialize** $\mu > 0$, β , $v = \alpha$. Set $1 < \lambda < 2$.
- 2: **while** not converge **do**
- 3: Update α and v by:
- 4: $\min_{\alpha^\top \mathbf{1} = 1, \alpha \geq 0} (\alpha^\top \mathcal{A} \alpha - 2b^\top \alpha) + \frac{\mu}{2} \|(\alpha - v) + \frac{\beta}{\mu}\|_F^2 \triangleright$ Eq. (33)
- 5: Update β by $\beta = \beta + \mu(\alpha - v)$
- 6: Update μ by $\mu = \lambda \mu$
- 7: **end while**

program over the simplex. Let $\mathbf{T} = \tilde{\mathbf{Y}} \mathbf{H}^\top$, the objective is

$$\left\| \sum_v \alpha^{(v)} \mathbf{K}^{(v)} - \mathbf{T} \right\|_F^2 \quad (29)$$

Then, it expands (29) into quadratic form:

$$\begin{aligned} \left\| \sum_v \alpha^{(v)} \mathbf{K}^{(v)} - \mathbf{T} \right\|_F^2 &= \sum_{v,w} \alpha^{(v)} \alpha^{(w)} \text{Tr}(\mathbf{K}^{(v)} \mathbf{K}^{(w)}) \\ &\quad - 2 \sum_v \alpha^{(v)} \text{Tr}(\mathbf{T}^\top \mathbf{K}^{(v)}) + \text{Tr}(\mathbf{T} \mathbf{T}^\top) \end{aligned} \quad (30)$$

where $\text{Tr}(\mathbf{T} \mathbf{T}^\top)$ is constant and can be omitted. Rather than forming the $n \times n$ matrices $\mathbf{K}^{(v)}$ explicitly, it can exploit:

$$\text{Tr}(\mathbf{K}^{(v)} \mathbf{K}^{(w)}) = \text{Tr}(\mathbf{B}^{(w)\top} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top} \mathbf{B}^{(w)}) \quad (31)$$

where $\mathbf{B}^{(v)} \in \mathbb{R}^{n \times m}$ ($m \ll n$). Thus, let Gram matrices $\mathbf{Q}_{ij} = \mathbf{B}^{(v)\top} \mathbf{B}^{(w)} \in \mathbb{R}^{m \times m}$ and obtain $\text{Tr}(\mathbf{K}^{(v)} \mathbf{K}^{(w)}) = \text{Tr}(\mathbf{Q}_{ij} \mathbf{Q}_{ij}^\top)$ with $O(m^2)$. Then, it form $\mathcal{A}$ and b by

$$\mathcal{A}_{ij} = \text{Tr}(\mathbf{Q}_{ij} \mathbf{Q}_{ij}^\top), \quad b_i = \text{Tr}(\mathbf{B}^{(i)\top} \mathbf{T}^\top \mathbf{B}^{(i)}) \quad (32)$$

This resulting sub-problem can be expressed as

$$\min_{\alpha} \alpha^\top \mathcal{A} \alpha - 2b^\top \alpha, \quad \text{s. t. } \alpha^\top \mathbf{1} = 1, \alpha \geq 0 \quad (33)$$

where $\mathcal{A}$ is a positive semi-definite matrix. Equation (33) is a convex Quadratic Programming (QP) problem, which can be updated by Algorithm 4. The cost to build the quadratic term and linear term is $O(V^2 n m^2 + V n m c)$.

D. Computational Complexity

The overall cost of ESADC includes an initialization stage and t alternating iterations, each of which scales linearly with the number of samples n . During initialization, K -means is run r times to select m anchors and constructs the $n \times m$ affinity matrix in $O(r n m d)$. The computation of $\mathbf{A}^\top \mathbf{A}$ takes $O(n m^2)$, raising it to the first V powers costs $O(V m^3)$, and multiplying each power back by $\mathbf{A}$ requires $O(V n m^2)$.

In each of the t alternating iterations, updating $\mathbf{H}$ involves projecting the current pseudo labels and aggregating across all V orders, which cost $O(V n m c)$, followed by a compact SVD

in $O(nc^2)$. Updating $\mathbf{Y}$ involves computing the clustering gain matrix in $O(Vnmc)$, followed by row-wise maximization in $O(nc^2)$. Updating α requires computing V^2 pairwise Gram matrices of size $m \times m$ with cost $O(V^2nm^2)$, and solving a small-scale quadratic program with negligible cost $O(V^3)$. Altogether, each iteration runs in

$$O(V^2nm^2 + Vnmc + nc^2 + nc^2 + V^3) \approx O(n) \quad (34)$$

where m , c , and V are typically much smaller than n . Therefore, ESADC scales linearly with dataset size.

E. Convergence Analysis

Theorem 1: It can be shown that ESADC's block-coordinate updates (Algorithms 2–4; $\mathbf{H}$, $\mathbf{Y}$, and α) monotonically decrease the objective

$$\mathcal{L}(\alpha, \mathbf{Y}, \mathbf{H}) = \left\| \sum_{v=1}^V \alpha^{(v)} \mathbf{B}^{(v)} \mathbf{B}^{(v)\top} - \mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-\frac{1}{2}} \mathbf{H}^\top \right\|_F^2 \quad (35)$$

is non-increasing at each update. Since $\mathcal{L}$ is bounded below by zero, the sequence of iterates converges to a (local) optimum.

Proof: The proof can be three parts, corresponding to three sub-problems. It finally fuses them into overall convergence.

1) *Convergence of the $\mathbf{H}$ sub-problem:* With α , $\mathbf{Y}$ fixed, the $\mathbf{H}$ -update solves

$$\min_{\mathbf{H}^\top \mathbf{H} = \mathbf{I}} \left\| \mathbf{S}(\alpha) - \tilde{\mathbf{Y}} \mathbf{H}^\top \right\|_F^2 \quad (36)$$

where $\tilde{\mathbf{Y}} = \mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-1/2}$ and **Algorithm 2** computes closed-form SVD solution guarantees

$$\mathcal{L}(\alpha, \mathbf{Y}, \mathbf{H}^{(t+1)}) \leq \mathcal{L}(\alpha, \mathbf{Y}, \mathbf{H}^{(t)}) \quad (37)$$

2) *Convergence of the $\mathbf{Y}$ sub-problem:* With α and $\mathbf{H}$ fixed, updating $\mathbf{Y}$ maximizes the trace form

$$\text{Tr}(\mathbf{H}(\mathbf{Y}(\mathbf{Y}^\top \mathbf{Y})^{-1/2})^\top \mathbf{S}(\alpha)) \quad (38)$$

subject to $\mathbf{Y}$ being row-wise one-hot. **Algorithm 3** updates each row to the cluster with the highest gain, guaranteeing the Frobenius-norm error never increases.

$$\mathcal{L}(\alpha, \mathbf{Y}^{(t+1)}, \mathbf{H}) \leq \mathcal{L}(\alpha, \mathbf{Y}^{(t)}, \mathbf{H}) \quad (39)$$

Since the one-hot set is finite and $\mathcal{L} \geq 0$, the coordinate descent terminates at a local optimum in finite steps.

3) *Convergence of the α sub-problem:* Holding $\mathbf{Y}$ and $\mathbf{H}$ fixed, updating α reduces to convex quadratic program

$$\min_{\alpha^\top \mathbf{1} = 1, \alpha \geq 0} \left\| \sum_{v=1}^V \alpha^{(v)} \mathbf{K}^{(v)} - \tilde{\mathbf{Y}} \mathbf{H}^\top \right\|_F^2 \quad (40)$$

Algorithm 4 finds a unique optimal α on the simplex, thus reducing objective:

$$\mathcal{L}(\alpha^{(t+1)}, \mathbf{Y}, \mathbf{H}) \leq \mathcal{L}(\alpha^{(t)}, \mathbf{Y}, \mathbf{H}) \quad (41)$$

TABLE II

OVERVIEW OF THE EIGHT BENCHMARK DATASETS USED IN OUR EXPERIMENTS

Datasets	Samples	Features	Classes
Derma [38]	336	34	6
Control [39]	600	60	6
CoraOS [40]	1248	6737	4
NUS [41]	2400	64	12
CoraHA [42]	2708	1433	7
S.S [43]	12340	30642	8
Bio [44]	20000	18888	20
S.O. [44]	20000	22956	20

4) *Overall Convergence:* Chaining the three updates shows that a full iteration of OMAGC satisfies

$$\mathcal{L}(\alpha^{(t+1)}, \mathbf{Y}^{(t+1)}, \mathbf{H}^{(t+1)}) \leq \mathcal{L}(\alpha^{(t)}, \mathbf{Y}^{(t)}, \mathbf{H}^{(t)}) \quad (42)$$

Since $\mathcal{L}$ is nonnegative and lower-bounded by zero, the sequence of objectives converges, and the corresponding variables converge to a local optimal point of the joint problem.

V. EXPERIMENTS

A. Datasets

To demonstrate the effectiveness of ESADC, we conducted comparative experiments on eight widely used machine learning benchmark datasets of varying sizes, including three normal-scale datasets Dermatology (Derma) [38], Control [39], CoraOS [40], and five large-scale datasets NUS [41], CoraHA [42], SearchSP (S.S.) [43], Biomedical (Bio) [44], and StackOF (S.O.) [44] to assess its clustering performance. Table II provides detailed information on each dataset. Moreover, we employ five evaluation metrics to assess clustering quality, including accuracy (ACC), normalized mutual information (NMI), purity, the Adjusted Rand Index (ARI), and the F-score. All experiments were conducted on a single desktop computer equipped with an Intel Core Ultra 7 155H processor running at 1.40 GHz, 256 GB of RAM, and MATLAB version 2024b, which ensures identical hardware and software.

For ESADC, the main hyperparameters include the anchor number m , graph order V , neighborhood size k , and bandwidth parameter σ . Unless otherwise specified, these parameters are selected by grid search on each dataset. In our experiments, V is searched in $\{2, 3, 4, 5, 6\}$ and σ is searched in $\{0.1, 0.2, 1, 2, 3\}$, while k is fixed to 5, following the common local-connectivity setting in anchor graph construction. Finally, the graph order is set to $V = 3$, and the anchor number m is chosen according to the dataset scale and further analyzed in Section V-D. To improve the stability of anchor selection, K -means is run 10 times with different random initializations, and the final anchors are obtained by averaging the anchor sets produced across these runs.

B. Baselines

To ensure an unbiased comparison, each baseline in this paper was carefully fine-tuned to match its original code or published version. We evaluate eight recent state-of-the-art methods with their best parameter combinations, including six traditional

clustering approaches and two deep learning-based models. These baselines follow different design strategies, particularly in how structural information is modeled and utilized for clustering. Traditional methods typically rely on explicit affinity construction, while deep clustering methods integrate representation learning and clustering by end-to-end network training. These differences are important for understanding the performance observed in the experimental results. The details of these baselines are as follows:

FSSC [45] selects anchors via BKHK, constructs a bipartite graph for self-supervised optimization, and then diffuses labels, delivering high-speed semi-supervised clustering.

NEWK [46] reframes K -means as a trace maximization problem, eliminating per-iteration centroid updates and reducing the variable via an iteratively reweighted scheme.

RSOCF [11] enhances concept factorization by enforcing orthogonal factors, directly deriving sample labels, and integrating a correntropy term to robustly handle complex noise.

AROCF [47] treats one factor matrix as an orthogonal cluster indicator and uses a compact anchor graph to approximate the full graph, reducing complexity.

JSSC [48] introduces joint sparse subspace clustering by adding an $\ell_{2,1}$ -norm for global sparsity and solves it via proximal alternating minimization.

MCHBG [49] proposes a high-order bipartite-graph fusion framework that enriches clustering with higher-order constructs while controlling complexity.

GJRD [50] uses a deep autoencoder with a differentiable Jensen–Rényi divergence estimator to enforce cluster separation, compactness, and orthogonality.

ICGCN [51] integrates contrastive learning and multi-scale GCNs on a CNN backbone, using a KL-based loss for adaptive feature fusion and state-of-the-art accuracy.

C. Clustering Performance

To provide a more intuitive manifestation of ESADC’s clustering performance, Fig. 3 presents t-SNE visualizations on four contrasting datasets: the normal-scale Derma and Control set, and the larger CoraHA and S.S. set. Although occasional minor imperfections can be observed, such as a few points lying near cluster borders, ESADC maintains clear boundary definitions between classes in each case. These results show that ESADC can achieve high clustering fidelity on small datasets and keep stable performance on large-scale.

We also evaluate ESADC against baselines on eight real-world datasets. Table III summarizes clustering results of each algorithm, and Table IV reports their corresponding run-times. In both, the highest and second-highest are bold and underlined, and “/” marks out-of-memory (Table III) or a runtime exceeding one hour (Table IV).

Table III shows that ESADC consistently leads or ties for the best performance across eight benchmark datasets and all five metrics (Accuracy, NMI, Purity, ARI, and F-score). On small datasets like Derma, it achieves 97.71% Accuracy (vs. 96.62%) and raises ARI from 92.29% to 93.29%, highlighting its ability to separate clusters even with limited samples. On

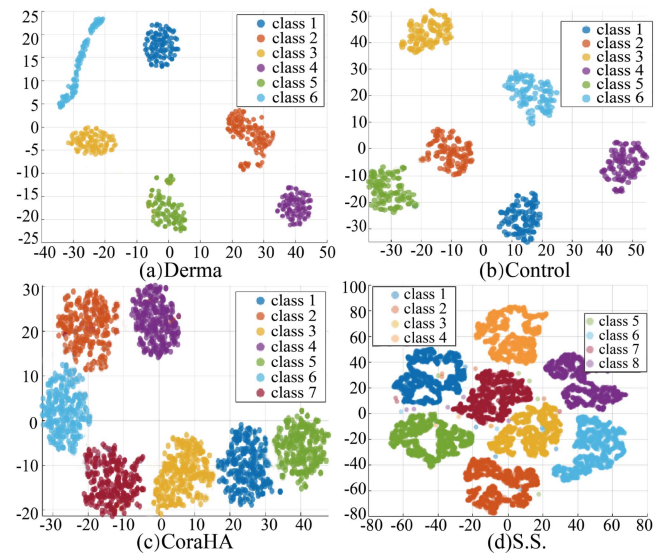


Fig. 3. t-SNE visualization of ESADC on four typical datasets: (a) Derma (normal-scale dataset); (b) Control (normal-scale dataset); (c) CoraHA (large-scale dataset); (d) S.S. (large-scale dataset).

medium-scale experiments such as Control and CoraOS, ESADC attains 96.04% (vs. 90.51%) and 53.61% (vs. 44.18%) Accuracy, with Purity climbing to 96.58% and 60.59%, respectively. On the mid-scale CoraHA dataset, it improves Accuracy to 49.00% (vs. 47.75%) and Purity to 52.75% (vs. 50.50%), demonstrating robustness even with a large feature set. Even at the very largest scales, such as S.S. (44.62% vs. 32.75%), Bio (19.89% vs. 19.78%), and S.O. (20.80% vs. 18.87%), ESADC remains clearly ahead. These gains, achieved without the heavy training and inference costs associated with deep models like GJRD and ICGCN, stem from ESADC’s adaptive multi-order anchor graph fusion and direct discrete clustering, which integrate local affinities with global topology to deliver robust and scalable performance.

Table IV reports the running time of ESADC and eight state-of-the-art methods across nine datasets, measured in seconds. Even on the smallest tests like Derma and Control, ESADC completes in just 0.0206 s and 0.0225 s, cutting the next fastest methods (0.0328 s and 0.0438 s) by over 40%. As the dataset size grows, its linear multi-order graph construction and one-step discrete clustering keep it efficient. On NUS, ESADC finishes in 0.3114 s versus 1.8470 s for the next best method (RSOCF). On CoraHA in 0.4949 s compared with 0.6730 s from NEWKmeans. On medium-scale experiments like CoraOS, it runs in 1.9997 s, narrowly edging out FSSC’s 2.0031 s. Crucially, on the largest benchmarks such as S.S., Bio, and S.O., ESADC requires only 123.8762 s, 169.6785 s, and 167.9104 s, respectively, dramatically lower than the deep-model baselines. Compared with deep clustering baselines, ESADC consistently achieves competitive or superior performance while requiring much lower computational cost. This advantage stems not only from its efficient anchor-based formulation, but also from its explicit modeling of structural information through multi-order anchor graph construction and adaptive fusion. By

TABLE III
COMPARISON RESULTS OF ESADC AND SOTA METHODS (%)

Metrics	Datasets	FSSC	NEWK	RSOCF	AROCF	JSSC	MCHBG	GJRD	ICGCN	ESADC
ACC	Derma	84.97±0.33	89.73±2.33	93.41±1.06	95.60±1.61	84.92±0.46	95.63±0.72	<u>96.62±0.54</u>	95.81±1.27	97.71±0.67
	Control	56.00±1.46	56.83±1.46	56.11±0.98	57.24±1.34	38.85±0.82	57.67±1.08	<u>90.51±1.77</u>	89.67±1.03	96.04±1.58
	CoraOS	43.74±0.83	43.87±1.27	43.65±1.06	43.11±1.14	43.88±1.40	40.32±0.10	<u>44.18±1.21</u>	44.07±0.98	53.61±1.64
	NUS	14.78±1.21	16.43±1.49	17.38±0.36	16.61±0.96	<u>19.70±1.21</u>	11.25±0.04	19.51±1.16	18.88±0.89	20.68±0.89
	CoraHA	41.25±0.49	32.75±0.49	40.88±1.12	41.64±1.67	41.30±1.88	47.75±1.75	42.04±1.15	41.64±1.02	49.00±1.98
	S.S.	21.03±0.81	21.56±0.87	30.90±2.61	30.75±3.16	25.88±2.13	23.61±1.32	<u>32.75±0.21</u>	32.12±1.11	44.62±1.89
	Bio	8.96±1.10	7.64±0.34	<u>19.78±1.25</u>	18.54±1.24	16.51±0.72	18.25±1.79	19.63±1.24	18.97±1.31	19.89±0.77
	S.O.	10.37±0.79	6.79±0.32	15.55±1.11	15.22±0.81	18.25±0.89	16.11±0.62	<u>18.87±0.62</u>	18.12±0.97	20.80±1.10
NMI	Derma	80.21±0.47	89.68±0.46	90.23±0.45	91.01±0.43	91.55±0.73	90.62±0.89	72.67±0.31	<u>91.59±1.70</u>	92.69±0.79
	Control	60.37±0.49	74.21±0.48	74.56±0.43	75.23±0.44	<u>78.54±1.02</u>	76.64±1.07	41.51±0.20	69.35±1.26	92.94±2.34
	CoraOS	13.44±1.19	12.65±0.99	14.73±1.22	11.68±1.11	2.66±0.90	<u>16.94±1.46</u>	13.51±0.89	9.66±0.97	17.08±1.29
	NUS	12.37±0.45	<u>22.46±0.47</u>	22.34±0.52	19.47±0.51	20.87±0.94	22.17±0.81	8.66±0.51	7.10±0.02	22.68±0.36
	CoraHA	0.73±0.43	15.57±0.49	14.36±0.47	13.45±0.50	<u>16.84±1.23</u>	15.11±0.23	6.72±1.14	14.31±0.13	18.32±0.73
	S.S.	26.54±0.46	<u>37.91±0.48</u>	34.21±0.45	35.24±0.47	37.66±1.81	35.01±0.78	8.49±0.11	14.97±0.49	39.72±0.75
	Bio	30.31±0.48	<u>42.99±0.43</u>	39.21±0.43	38.89±0.44	40.92±1.01	40.42±0.66	7.05±0.61	9.62±0.08	43.12±0.52
	S.O.	35.77±0.42	46.32±0.44	47.24±0.34	48.16±0.50	50.91±1.17	50.06±0.91	8.72±0.31	11.92±0.89	<u>49.22±1.44</u>
Purity	Derma	84.97±0.46	92.02±0.47	93.12±0.43	93.45±0.42	95.14±1.09	93.67±0.42	84.92±1.63	<u>95.63±1.38</u>	96.04±0.67
	Control	56.17±0.49	66.67±0.54	71.32±0.45	71.45±0.47	<u>89.67±0.81</u>	88.91±1.02	47.83±0.38	64.67±1.61	96.58±1.58
	CoraOS	43.74±0.83	43.87±1.27	43.65±1.06	43.11±1.14	43.88±1.40	40.32±0.10	<u>44.18±1.21</u>	44.07±0.98	60.59±1.64
	NUS	48.85±1.27	52.25±1.91	18.24±0.47	56.93±1.29	51.84±0.87	50.42±0.90	21.85±0.76	19.63±0.13	52.36±0.54
	CoraHA	41.75±0.44	49.45±0.46	48.28±0.54	47.60±0.56	48.75±0.91	41.87±1.23	42.10±0.77	<u>50.50±0.77</u>	52.75±0.38
	S.S.	<u>47.80±1.32</u>	46.13±1.93	33.70±3.63	33.25±3.37	32.54±0.62	39.64±0.92	23.11±0.49	17.89±0.39	52.69±1.03
	Bio	28.23±1.58	28.91±0.95	20.93±1.08	<u>43.67±0.85</u>	40.61±1.59	40.12±1.04	18.44±0.30	23.12±0.85	44.17±1.45
	S.O.	29.70±0.69	<u>33.23±1.96</u>	16.56±1.03	16.15±0.96	31.74±1.65	28.46±0.96	16.95±0.49	19.48±0.16	42.11±1.12
ARI	Derma	80.93±2.21	86.50±1.62	85.53±2.21	87.45±2.31	92.15±0.24	90.36±1.01	74.44±0.68	<u>92.29±1.75</u>	93.29±0.99
	Control	49.39±0.13	61.72±1.44	58.99±1.14	60.72±1.21	<u>78.27±0.96</u>	76.91±0.89	24.37±1.57	58.68±0.06	92.33±3.36
	CoraOS	5.20±0.86	5.28±0.90	5.18±0.87	5.12±0.85	0.20±1.84	0.10±0.25	<u>9.46±0.92</u>	7.38±0.90	10.44±0.03
	NUS	41.36±1.83	43.39±1.98	44.51±2.06	<u>45.72±1.82</u>	42.24±1.61	41.62±1.05	3.35±1.12	17.71±1.30	46.11±0.68
	CoraHA	0.38±0.12	<u>8.98±0.15</u>	6.42±0.55	7.02±0.62	8.56±2.15	8.07±1.12	0.30±0.69	2.75±0.68	17.08±0.80
	S.S.	40.94±1.50	42.69±1.78	39.78±1.24	41.55±1.46	53.88±1.21	55.92±0.83	7.30±0.24	3.76±0.17	53.95±0.14
	Bio	27.23±1.44	24.56±1.35	28.44±1.17	30.00±1.09	<u>38.67±0.43</u>	36.32±0.96	5.42±0.98	12.32±0.16	38.72±0.69
	S.O.	22.71±0.78	23.41±1.64	24.11±1.33	23.07±1.12	24.99±0.64	<u>25.87±0.63</u>	3.74±0.83	6.69±0.96	26.79±0.16
F-score	Derma	84.87±1.22	89.24±1.53	91.22±1.61	92.65±1.18	92.91±0.79	91.44±0.59	28.83±0.27	<u>93.82±1.97</u>	94.63±0.80
	Control	59.50±0.54	69.44±0.43	61.43±0.79	64.56±0.95	<u>79.77±0.61</u>	71.22±1.32	19.62±0.87	67.12±0.17	93.60±2.80
	CoraOS	44.84±1.16	44.92±1.20	44.78±1.17	44.65±1.14	33.86±1.29	30.00±1.09	<u>45.12±1.21</u>	45.07±1.19	48.70±0.92
	NUS	49.37±2.32	50.64±2.12	53.87±2.22	54.23±2.31	52.66±1.43	51.27±1.01	8.36±1.24	11.97±0.82	<u>53.91±0.82</u>
	CoraHA	<u>41.32±1.07</u>	27.31±0.10	38.89±0.14	40.22±0.62	41.09±0.87	40.73±1.51	2.07±1.11	39.08±0.38	45.81±0.51
	S.S.	45.95±1.85	46.34±1.73	<u>48.72±1.67</u>	23.84±1.44	42.69±1.23	40.14±0.76	7.30±0.24	5.06±0.14	49.62±0.79
	Bio	33.24±1.49	35.32±1.61	36.43±1.22	37.74±1.06	<u>39.11±1.53</u>	34.87±1.04	3.22±0.47	9.16±0.81	39.31±0.47
	S.O.	28.75±1.22	32.29±1.82	31.12±1.57	8.24±0.43	<u>32.75±0.87</u>	30.99±1.06	4.23±0.73	3.05±0.75	35.24±0.69

integrating complementary local-to-global manifold information at the graph level, ESADC provides a richer and more controllable structural representation. In contrast, deep clustering methods typically acquire such structural cues implicitly through extensive end-to-end training. These results indicate that explicit multi-scale structural modeling can be a highly effective and efficient strategy for clustering. Overall, these results clearly show that ESADC consistently achieves superior clustering quality while simultaneously reducing computational overhead across datasets of varying scales.

D. Parameter Sensitivities

To verify that the fused multi-order graph reveals its intrinsic structure, Fig. 4 visualizes proximity matrices at different fusion orders on four typical datasets. Almost in all typical datasets

of Fig. 4, the 1st- and 2nd-order fusions remain overly sparse, failing to provide sufficient must-link information and thus losing essential structural cues. By contrast, the 3rd-order fusion recovers a clear block-diagonal pattern that best aligns with the ground-truth structure (cf. the rightmost ground-truth panel). As the fusion order increases beyond the 3rd, the block-diagonal clarity progressively degrades, and cluster boundaries become blurred. This deterioration stems from the exponential growth of multi-hop path counts, which introduces numerous spurious cross-cluster connections and causes originally compact clusters to interfere with one another. Overall, the figure demonstrates that a moderate fusion order (here, the 3rd) achieves the best trade-off between capturing higher-order relations and avoiding over-smoothing from excessive multi-hop connectivity. Hence, only by balancing local and global perspectives, that is, by fusing low-order and suitably chosen high-order information, can both

TABLE IV
RUNNING TIME OF ESADC AND THE STATE-OF-THE-ART METHODS (SECONDS)

Datasets	FSSC	NEWK	RSOCF	AROCF	JSSC	MCHBG	GJRD	ICGCN	ESADC
Derma	0.0714	<u>0.0328</u>	0.6457	0.6542	2.0282	1.0094	7.5436	6.2561	0.0206
Control	0.0993	<u>0.0438</u>	1.5463	0.8764	9.0273	3.8081	26.2311	20.1547	0.0225
CoraOS	<u>2.0031</u>	9.6786	10.2546	36.6321	2575.7192	365.1225	39.7367	33.7642	1.9997
NUS	8.7521	4.6121	<u>1.8470</u>	3.5461	213.6179	102.5773	30.5142	28.6354	0.3114
CoraHA	0.7433	<u>0.6730</u>	8.2713	5.3654	10.5053	20.2973	5.0654	3.9687	0.4949
S.S.	OM	4911.1150	11900.5547	<u>591.0000</u>	845.1497	10602.4278	OM	118632.354	123.8762
Bio	5879.5700	<u>602.7324</u>	4044.4690	6265.3287	5865.3086	7434.2684	9475.6571	7475.4532	169.6785
S.O.	12256.2587	11402.0000	5704.2485	<u>1265.8500</u>	4623.2915	8765.1663	35426.5478	29874.6431	167.9104

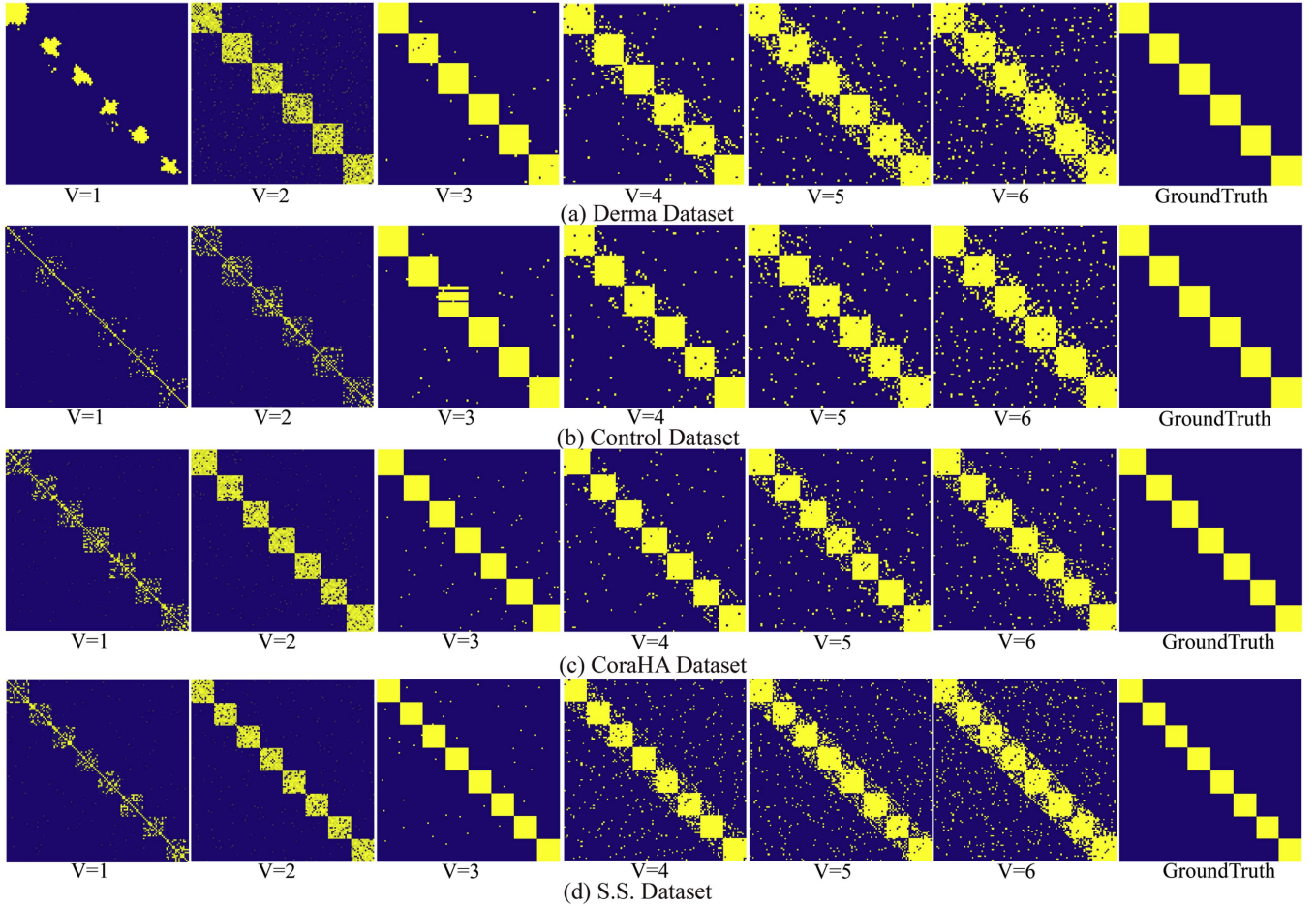


Fig. 4. Proximity matrices of various V -order fused graphs on four typical datasets: (a) Derma; (b) Control; (c) CoraHA; (d) S.S.

preserve fine-grained intra-cluster structure and maintain clear inter-cluster separation, thereby optimizing clustering performance.

Building on the above analysis, we next examine how the multi-order level V and the bandwidth σ jointly influence the performance. To assess their impact on clustering across datasets of varying sizes, we performed a comprehensive grid search over $V \in \{2, 3, 4, 5, 6\}$ and $\sigma \in \{0.1, 0.2, 1, 2, 3\}$, respectively. Fig. 5 displays how each (V, σ) combination affects ESADC's accuracy on eight representative benchmarks. Fig. 5 indicates that the ACC does not monotonically increase with larger V , rather, the optimal performance on every dataset occurs at

$V = 3$. This finding suggests that third-order fusion strikes the optimal balance between capturing essential deep interactions and avoiding the redundancy and noise introduced by excessively high orders. By contrast, higher orders beyond three begin to degrade performance, echoing our earlier observation that overly dense multi-hop connectivity blurs cluster boundaries. Moreover, the impact of the bandwidth σ varies across datasets. For example, $\sigma = 0.2$ obtains the highest accuracy on small-scale Derma while $\sigma = 2$ results in the best on large-scale Bio, which underscores the importance of tuning σ to the specific data distribution to fully leverage ESADC's clustering capabilities.

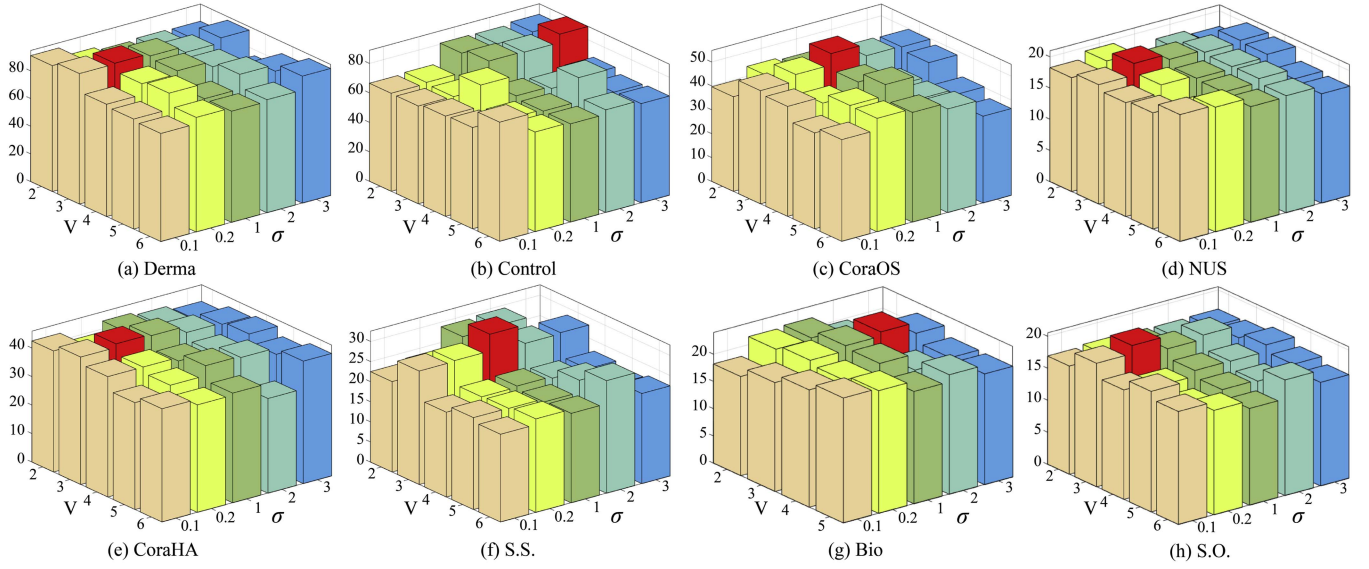


Fig. 5. Parameters of ESADC on eight datasets: (a) Derma; (b) Control; (c) CoraOS; (d) NUS; (e) CoraHA; (f) S.S.; (g) Bio; (h) S.O.

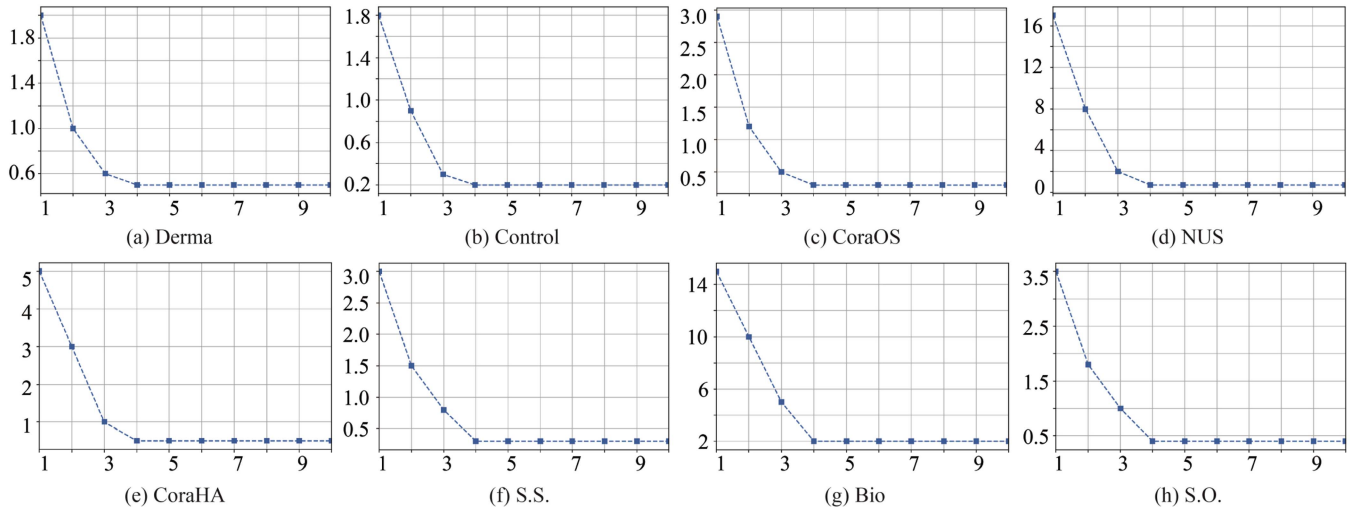


Fig. 6. Convergence curves of ESADC on: (a) Derma; (b) Control; (c) CoraOS; (d) NUS; (e) CoraHA; (f) S.S.; (g) Bio; (h) S.O.

To further analyze the sensitivity of ESADC to the anchor number m , we vary m while fixing all other parameters and report the corresponding clustering performance and running time on the representative small-scale dataset (Derma) and large-scale dataset (S.S.). It can be observed from Fig. 8 that the clustering performance improves rapidly when m increases from a small value, and then becomes stable once sufficient structural information has been captured. Further increasing m brings only a limited additional benefit. Meanwhile, Fig. 9 shows that the running time increases with m , with a more pronounced growth at larger anchor sizes. Thus, in ESADC, for small datasets such as Derma, the number of anchors is kept at a moderate level, since the performance becomes stable once the main structural information has been captured. For large-scale datasets such as S.S., a relatively larger anchor set is preferred to capture finer structural details.

To further assess the stability of anchor selection, we compare $K\text{-means}(1)$ and $K\text{-means}(10)$ on four representative datasets. $K\text{-means}(1)$ uses a single K-means run to generate anchors, while $K\text{-means}(10)$ runs K-means 10 times under different random initializations and derives the final anchors by averaging the anchor sets obtained from these runs. As shown in Table V, $K\text{-means}(10)$ consistently achieves better ACC, NMI, and ARI, while introducing only a minor increase in running time. These results confirm that repeated K-means initialization improves the robustness of anchor generation in ESADC without significantly affecting its efficiency.

E. Convergence Experiments

Fig. 6 depicts the convergence curves over eight typical datasets from small to large. As the plots demonstrate, the value

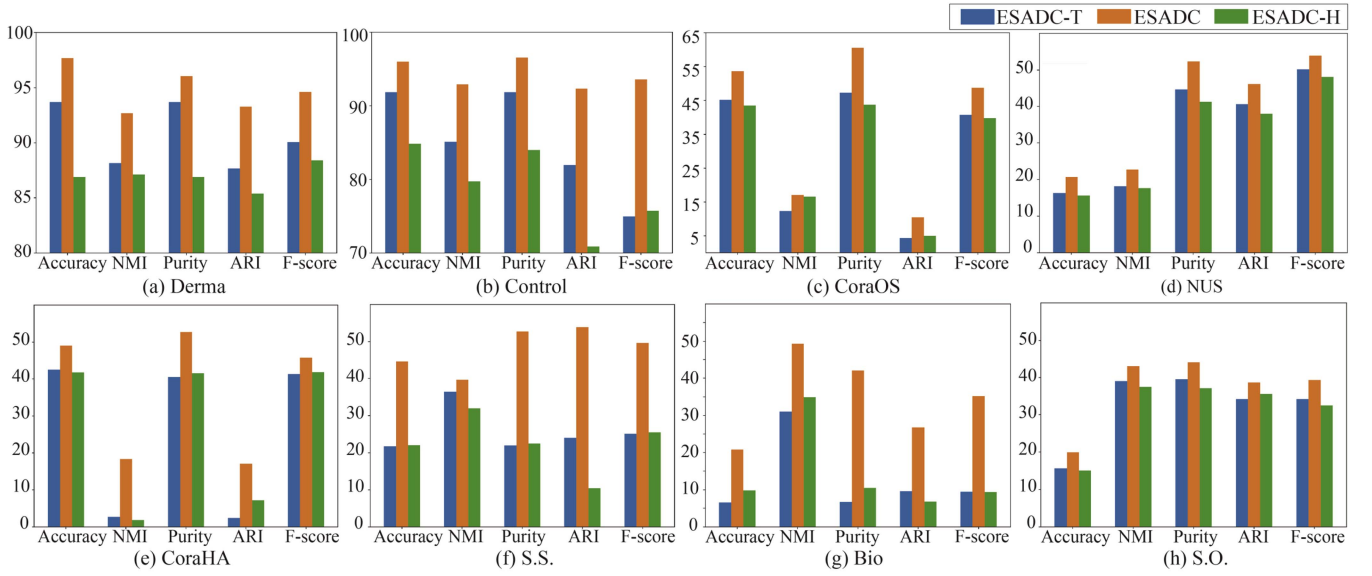


Fig. 7. Ablation study of ESADC, ESADC-T, and ESADC-H on: (a) Derma; (b) Control; (c) CoraOS; (d) NUS; (e) CoraHA; (f) S.S.; (g) Bio; (h) S.O.

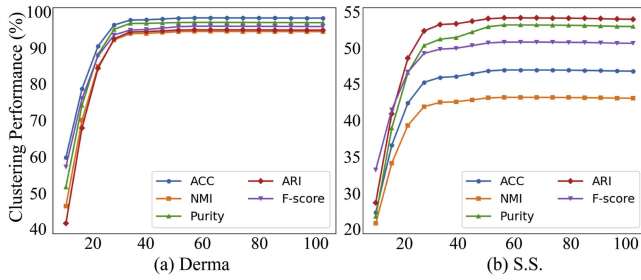


Fig. 8. Clustering results with different m on two representative datasets.

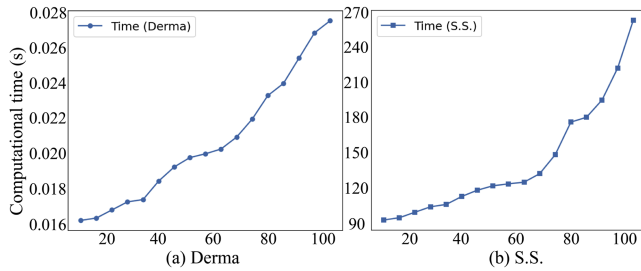


Fig. 9. Computational time with different m on two representative datasets.

TABLE V
COMPARISON OF INITIALIZATION BY K-MEANS(1) AND K-MEANS(10)

Metric	Method	Derma	Control	CoraHA	S.S.
ACC (%)	K-means(1)	97.12±0.86	94.88±2.11	47.83±2.37	42.91±2.54
	K-means(10)	97.71±0.67	96.04±1.58	49.00±1.98	44.62±1.89
NMI (%)	K-means(1)	91.94±1.02	90.73±2.88	17.41±1.05	38.46±1.18
	K-means(10)	92.69±0.79	92.94±2.34	18.32±0.73	39.72±0.75
ARI (%)	K-means(1)	92.48±1.21	90.81±3.94	15.92±1.14	52.31±0.39
	K-means(10)	93.29±0.99	92.33±3.36	17.08±0.80	53.95±0.14
Time (s)	K-means(1)	0.0189	0.0203	0.4628	118.7421
	K-means(10)	0.0206	0.0225	0.4949	123.8762

of the objective declines rapidly, reaching stability in as few as four steps. This swift convergence stems from the convexity of the subproblems, which accelerates their solution. These results also confirm that our alternating update scheme reliably drives the objective toward its minimum, enabling ESADC to cluster data both rapidly and dependably across diverse datasets.

F. Ablation Study

To validate the effectiveness of one-step discretization strategy and multi-order anchor graph fusion in clustering, we designed corresponding ablation experiments to isolate ESADC's core advantages. First, we replaced ESADC's one-step discretization with the conventional two-stage framework while preserving all other components unchanged, and refer to this variant as ESADC-T. Next, we substituted the multi-order anchor-graph fusion with a single high-order graph (chosen as the order that yields the highest accuracy for each dataset), again keeping all other modules identical, denoted ESADC-H. Fig. 7 shows the ablation study of ESADC, ESADC-T, and ESADC-H on six typical datasets. The experimental results demonstrate that ESADC model with both the one-step discretization strategy and multi-order anchor-graph fusion outperforms both ESADC-T and ESADC-H across all five clustering metrics (Accuracy, NMI, Purity, ARI, and F-score) on each of the six benchmark datasets. In every case, the margin of improvement is statistically significant, demonstrating that neither the conventional two-stage discretization nor a single high-order graph representation can match the combined strength of our dual innovations. Generally, these two modules address complementary challenges in graph-based clustering: the one-step discretization tightly couples embedding and partitioning to minimize error, and the multi-order fusion enriches the similarity structure to reveal true cluster geometry. Their synergy contributes to ESADC achieving superior performance in our experiments.

VI. CONCLUSION

The proposed ESADC integrates multi-order anchor graph fusion and one-step discrete optimization into a unified framework. It avoids the conventional two-stage pipeline and directly produces discrete cluster labels from the fused graph. Nevertheless, the proposed ESADC also has certain limitations. As a structure-driven clustering approach, it mainly exploits affinity relationships encoded by the constructed graphs, and may be less effective when the underlying structural information is weak or ambiguous. Moreover, ESADC does not explicitly incorporate additional semantic or contextual information beyond affinity modeling.

Beyond ESADC, multi-order anchor graph fusion offers a general and scalable paradigm for structure-aware learning, enabling unified modeling of data structures from local to global scales. This idea is potentially applicable to a broader range of large-scale graph-based learning problems, such as multi-view and semi-supervised learning. Future work may further explore task-driven fusion strategies to extend the applicability of this paradigm to more complex data settings.

REFERENCES

- [1] S. Ding, C. Li, X. Xu, L. Guo, L. Ding, and X. Wu, "Fast density peaks clustering algorithm based on approximate K-nearest neighbors," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 10, pp. 5878–5889, Oct. 2025.
- [2] Y. Gan et al., "Multi-view clustering via multi-stage fusion," *IEEE Trans. Multimedia*, vol. 27, pp. 4571–4583, 2025.
- [3] Z. Zhang, Y. Zhang, H. Tian, A. Martin, Z. Liu, and W. Ding, "A survey of evidential clustering: Definitions, methods, and applications," *Inf. Fusion*, vol. 115, 2025, Art. no. 102736.
- [4] Y. Zhang, S. Yan, L. Zhang, and B. Du, "Fast projected fuzzy clustering with anchor guidance for multimodal remote sensing imagery," *IEEE Trans. Image Process.*, vol. 33, pp. 4640–4653, 2024.
- [5] B. Yang, X. Zhang, J. Wu, F. Nie, F. Wang, and B. Chen, "Scalable min-max multi-view spectral clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 37, no. 5, pp. 2918–2931, May 2025.
- [6] Q. Liu, S. Chen, R. Jiang, and W. H. Wong, "Simultaneous deep generative modelling and clustering of single-cell genomic data," *Nature Mach. Intell.*, vol. 3, no. 6, pp. 536–544, 2021.
- [7] J. Gao et al., "Incomplete multimodal clustering via semidiscrete optimal transport for multimedia data mining in IoT," *ACM Trans. Multimedia Comput., Commun. Appl.*, vol. 20, no. 6, pp. 1–20, 2024.
- [8] S. Xia et al., "Ball k k-means: Fast adaptive clustering with no bounds," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 1, pp. 87–99, Jan. 2022.
- [9] J. Li and C. Li, "One-hot constrained symmetric nonnegative matrix factorization for image clustering," *Pattern Recognit.*, vol. 162, 2025, Art. no. 111427.
- [10] Y. Zhou, B. Yang, X. Zhang, and B. Chen, "Parameter-free discrete clustering via adaptive hypergraph fusion," *Inf. Sci.*, vol. 723, 2025, Art. no. 122677.
- [11] B. Yang et al., "Robust spectral embedded bilateral orthogonal concept factorization for clustering," *Pattern Recognit.*, vol. 150, 2024, Art. no. 110308.
- [12] Y. Liu et al., "Revisiting modularity maximization for graph clustering: A contrastive learning perspective," in *Proc. 30th ACM SIGKDD Conf. Knowl. Discov. Data Mining*, 2024, pp. 1968–1979.
- [13] F. Nie, J. Xue, W. Yu, and X. Li, "Fast clustering with anchor guidance," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 4, pp. 1898–1912, Apr. 2024.
- [14] Q. Qiang, B. Zhang, F. Wang, and F. Nie, "Fast multi-view discrete clustering with anchor graphs," in *Proc. AAAI Conf. Artif. Intell.*, 2021, pp. 9360–9367.
- [15] X. Fan, W. Yu, F. Nie, and X. Li, "Fast semi-supervised classification based on anchor graph," *Inf. Sci.*, vol. 699, 2025, Art. no. 121786.
- [16] M. Wang, W. Fu, S. Hao, D. Tao, and X. Wu, "Scalable semi-supervised learning by efficient anchor graph regularization," *IEEE Trans. Knowl. Data Eng.*, vol. 28, no. 7, pp. 1864–1877, Jul. 2016.
- [17] Z. Wang, X. Dai, P. Zhu, R. Wang, X. Li, and F. Nie, "Fast optimization of spectral embedding and improved spectral rotation," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 2, pp. 1515–1527, Feb. 2023.
- [18] R. Zhang, S. Hang, Z. Sun, F. Nie, R. Wang, and X. Li, "Anchor-based fast spectral ensemble clustering," *Inf. Fusion*, vol. 113, 2025, Art. no. 102587.
- [19] P. Li and O. Milenkovic, "Inhomogeneous hypergraph clustering with applications," in *Proc. Adv. Neural Inf. Process. Syst.*, 2017, pp. 2309–2319.
- [20] F. Li et al., "K-hyperedge medoids for clustering ensemble," in *Proc. AAAI Conf. Artif. Intell.*, 2025, pp. 18271–18278.
- [21] Z. Li, X. Yang, and C. Jiang, "Multi-view graph based hierarchical representation learning for money laundering group detection," *IEEE Trans. Inf. Forensics Secur.*, vol. 20, pp. 2035–2050, 2025.
- [22] C. Liu, J. Wen, Y. Xu, B. Zhang, L. Nie, and M. Zhang, "Reliable representation learning for incomplete multi-view missing multi-label classification," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 6, pp. 4940–4956, Jun. 2025.
- [23] J. Tang, M. Qu, M. Wang, M. Zhang, J. Yan, and Q. Mei, "LINE: Large-scale information network embedding," in *Proc. 24th Int. Conf. World Wide Web*, 2015, pp. 1067–1077.
- [24] Z. Zhao et al., "Enhancing clustering performance with tensorized high-order bipartite graphs: A structured graph learning approach," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 35, no. 3, pp. 2616–2631, Mar. 2025.
- [25] B. Jiang et al., "Scalable fuzzy clustering with collaborative structure learning and preservation," *IEEE Trans. Fuzzy Syst.*, vol. 33, no. 9, pp. 3047–3060, Sep. 2025.
- [26] J. Yang et al., "Multi-view clustering via multi-granularity ensemble," in *Proc. 34th Int. Joint Conf. Artif. Intell.*, 2025, pp. 6794–6802.
- [27] B. Jiang et al., "Semi-supervised multi-view feature selection with adaptive similarity fusion and learning," *Pattern Recognit.*, vol. 159, 2025, Art. no. 111159.
- [28] B. Jiang et al., "Scalable semi-supervised learning with discriminative label propagation and correction," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 48, no. 6, pp. 6156–6173, Jun. 2026.
- [29] N. Wang, Z. Cui, A. Li, Y. Lu, R. Wang, and F. Nie, "Structured doubly stochastic graph-based clustering," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 6, pp. 11064–11077, Jun. 2025.
- [30] Y. Pang, J. Xie, F. Nie, and X. Li, "Spectral clustering by joint spectral embedding and spectral rotation," *IEEE Trans. Cybern.*, vol. 50, no. 1, pp. 247–258, Jan. 2020.
- [31] Y. Han, L. Zhu, Z. Cheng, J. Li, and X. Liu, "Discrete optimal graph clustering," *IEEE Trans. Cybern.*, vol. 50, no. 4, pp. 1697–1710, Apr. 2020.
- [32] L. Huang, L. Gan, and B. W.-K. Ling, "A unified optimization model of feature extraction and clustering for spike sorting," *IEEE Trans. Neural Syst. Rehabil. Eng.*, vol. 29, pp. 750–759, 2021.
- [33] H. Wang, Y. Yang, and B. Liu, "GMC: Graph-based multi-view clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 32, no. 6, pp. 1116–1129, Jun. 2020.
- [34] X. Qiao, M. Poggi, X. Wei, P. Deng, Y. Zhou, and S. Mattoccia, "Learnable fractional reaction-diffusion dynamics for under-display ToF imaging and beyond," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, 2025, pp. 6080–6090.
- [35] C. Tang, Z. Li, J. Wang, X. Liu, W. Zhang, and E. Zhu, "Unified one-step multi-view spectral clustering," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 6, pp. 6449–6460, Jun. 2023.
- [36] J. Wang, C. Tang, Z. Wan, W. Zhang, K. Sun, and A. Y. Zomaya, "Efficient and effective one-step multiview clustering," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 35, no. 9, pp. 12224–12235, Sep. 2024.
- [37] X. Wan et al., "One-step multi-view clustering with diverse representation," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 36, no. 3, pp. 5774–5786, Mar. 2025.
- [38] N. Ilter and H. Guvenir, "Dermatology," [Dataset]. *UCI Mach. Learn. Repository*, Univ. California, Irvine, Irvine, CA, USA, 1998, doi: [10.24432/C5FK5P](https://doi.org/10.24432/C5FK5P).
- [39] R. Alcock, "Synthetic control chart time series," [Dataset]. *UCI Mach. Learn. Repository*, Univ. California, Irvine, Irvine, CA, USA, 1999, doi: [10.24432/C59G75](https://doi.org/10.24432/C59G75).
- [40] P. Sen, G. Namata, M. Bilgic, L. Getoor, B. Gallagher, and T. Eliassi-Rad, "Collective classification in network data," *AI Mag.*, vol. 29, no. 3, pp. 93–93, 2008.
- [41] T.-S. Chua, J. Tang, R. Hong, H. Li, Z. Luo, and Y. Zheng, "NUS-WIDE: A real-world web image database from national university of Singapore," in *Proc. ACM Int. Conf. Image Video Retrieval*, 2009, pp. 1–9.

- [42] A. K. McCallum, K. Nigam, J. Rennie, and K. Seymore, "Automating the construction of internet portals with machine learning," *Inf. Retrieval*, vol. 3, pp. 127–163, 2000.
- [43] X.-H. Phan, L.-M. Nguyen, and S. Horiguchi, "Learning to classify short and sparse text & web with hidden topics from large-scale data collections," in *Proc. 17th Int. Conf. World Wide Web*, 2008, pp. 91–100.
- [44] J. Xu, B. Xu, P. Wang, S. Zheng, G. Tian, and J. Zhao, "Self-taught convolutional neural networks for short text clustering," *Neural Netw.*, vol. 88, pp. 22–31, 2017.
- [45] J. Wang, Z. Ma, F. Nie, and X. Li, "Fast self-supervised clustering with anchor graph," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 33, no. 9, pp. 4199–4212, Sep. 2022.
- [46] F. Nie, Z. Li, R. Wang, and X. Li, "An effective and efficient algorithm for k-means clustering with new formulation," *IEEE Trans. Knowl. Data Eng.*, vol. 35, no. 4, pp. 3433–3443, Apr. 2023.
- [47] B. Yang, Z. Xue, J. Wu, X. Zhang, F. Nie, and B. Chen, "Anchor-graph regularized orthogonal concept factorization for document clustering," *Neurocomputing*, vol. 573, 2024, Art. no. 127173.
- [48] Y. Zhu, X. Xiu, W. Liu, and C. Yin, "Joint sparse subspace clustering via fast $l(2,0)$ -norm constrained optimization," *Expert Syst. Appl.*, vol. 265, 2025, Art. no. 125845.
- [49] Z. Zhao, T. Wang, H. Xin, R. Wang, and F. Nie, "Multi-view clustering via high-order bipartite graph fusion," *Inf. Fusion*, vol. 113, 2025, Art. no. 102630.
- [50] M. Lu, L. Xing, and B. Chen, "Measuring generalized divergence for multiple distributions with application to deep clustering," *Pattern Recognit.*, vol. 157, 2025, Art. no. 110864.
- [51] Y. Xu, D. Huang, C.-D. Wang, and J.-H. Lai, "Deep image clustering with contrastive learning and multi-scale graph convolutional networks," *Pattern Recognit.*, vol. 146, 2024, Art. no. 110065.



Ben Yang (Member, IEEE) received the BS degree in automation science and technology and the PhD degree in control science and engineering from Xi'an Jiaotong University, Shaanxi, China, in 2016 and 2023, respectively. He visited the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore, from 2021 to 2022. He is currently an associate professor with the Institute of Artificial Intelligence and Robotics, College of Artificial Intelligence, Xi'an Jiaotong University. His research interests include data mining and machine learning.



Xuetao Zhang (Member, IEEE) received the BS degree in information engineering, the MS degree, and the PhD degree in pattern recognition and intelligent system from Xi'an Jiaotong University, China, in 2003, 2006, and 2012, respectively. He visited the Department of Brain and Cognitive Sciences, Massachusetts Institute of Technology from 2009 to 2010. He is currently a professor with the Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University. His research includes computer vision and machine learning.



Yu Zhou received the PhD degree from the Huazhong University of Science and Technology, Wuhan, China, in 2022. He was also a visiting scholar with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. He is currently an assistant professor with the School of Information Engineering, Zhongnan University of Economics and Law. His research interests include computer vision, deep learning, quaternion neural networks, and spectral clustering.



Haoxin Wu is currently working toward the BS degree with the School of Information Engineering, Zhongnan University of Economics and Law, Wuhan, China. His research interests include computer vision and deep learning, with a focus on spectral clustering and its application to data mining. He is particularly interested in quaternion neural networks for multi-channel feature representation and clustering methods for complex data.



Feiping Nie (Senior Member, IEEE) received the PhD degree in computer science from Tsinghua University, China, in 2009. He is currently a full professor with Northwestern Polytechnical University, China. He has authored or coauthored more than 100 papers in journals and conferences such as *IEEE Transactions on Pattern Analysis and Machine Intelligence*, *International Journal of Computer Vision*, *IEEE Transactions on Image Processing*, *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Knowledge and Data Engineering*, *ICML*, *NIPS*, *KDD*, *IJCAI*, *AAAI*, *ICCV*, *CVPR*, and *ACM MM*. His research interests include machine learning and its applications, such as pattern recognition, data mining, computer vision, image processing, and information retrieval. His papers have been cited more than 36000 times and the H-index is 102. He is an associate editor or PC member for several prestigious journals and conferences in the related fields.



Badong Chen (Senior Member, IEEE) received the PhD degree in computer science and technology from Tsinghua University, Beijing, China, in 2008. He is currently a professor with the Institute of Artificial Intelligence and Robotics, Xi'an Jiaotong University, Xi'an, China. He has authored or co-authored more than 200 articles in various journals and conference proceedings. His research interests include signal processing, machine learning, artificial intelligence and robotics. He won the 2022 Outstanding Paper Award of *IEEE Transactions on Cognitive and Developmental Systems*. He is a technical committee member of IEEE SPS Machine Learning for Signal Processing. He is/was as an associate editor for several international journals including *IEEE Transactions on Neural Networks and Learning Systems*, *IEEE Transactions on Cognitive and Developmental Systems*, *Neural Networks*, and *Journal of The Franklin Institute*. He was a PC or SPC member for top-ranked conferences including *IJCAI*, *AAAI*, and *UAI*. He has been selected as a general co-chair of IEEE International Workshop on Machine Learning for Signal Processing.